

BE DOING MORE WITH LESS

Abundant energy... feels fishy right? The sun delivering ten thousand times more energy than we need. Learning curves making clean alternatives super cheap. Fossil fuel simply displaced by something better. No running out of raw materials anymore. It all sounds too good to be true. It *feels* too good to be true. I know that feeling, believe you me.

I'm Dutch. We have a Calvinist tradition that runs deep, a conviction that nothing good comes without suffering. "By the sweat of your brow you shall eat bread," Genesis tells us. No easy guilt-free flourishing for us. No having your cake and eat it too. No such thing as a free lunch.

I try hard to be sustainable. I drive electric. I build a house from wood. It produces more energy than I use. I avoid flying. Eat no meat. I really try. But I still feel guilty. Because it's actually a very *nice* wooden house. A very *nice* electric car. And my wife's vegan cooking is *wonderful*. So it feels suspiciously easy. *Too* easy. Why would I have it so good when so many others have it so bad?

That's the kind of person I am. So take it from me: I take the objections to abundant energy very seriously. Both rationally and emotionally. I've spent most of my life wrestling with them, and I'm still wrestling. I can't help it. It's also what initially attracted me to the report *The Limits of Growth* as described in chapter 1.1 and later the degrowth movement. They put things I was taught to care about first: helping people in poor countries, staying within planetary boundaries, not chasing after endless growth on a finite planet. That felt trustworthy. Morally serious.

Some readers will recognize this feeling. Others might think I'm strange. But at least I hope you all believe me when I say I take the objections to abundance very seriously. Nevertheless, I was wrong.

8.1 The Things We Really Need to Degrow

Degrowth appealed to my guilt. It told me that my suspicion of comfort was correct, that abundance itself was the problem, that the only ethical path was less. Consume less, produce less, want less. Share the shrinking pie more fairly. And degrowth is right that we are already exceeding planetary boundaries while far too many people live in abject poverty and we should do something about all that.

But the solution they offer doesn't survive contact with reality. Austerity combined with redistribution. Managed decline for the wealthy nations. These are political nonstarters. You are asking billions of people to vote for decline, and democracies don't vote for decline. Dictators don't choose it either, by the way. You might get a few percentage points of voluntary reduction, but you will never get the factor ten or more that we need. It's just magical thinking that feels good when you're on a guilt trip.

It also doesn't help the planet. Even if you somehow achieved this unprecedented redistribution, the planet wouldn't notice. You'd have the same total energy use, just distributed differently. The same carbon emissions. The same material extraction. The planet doesn't care who burns the coal. To really make an impact through reduction, we would have to use radically less energy globally, and in any realistic scenario that would cause suffering on a scale I find unacceptable. Simpler technology doesn't help either. I discovered that per person, hunter-gatherers needed an area of between 20 and a whopping 1,000 square kilometres. ([NatGeo Education, n.d.](#)); ([Britannica, n.d.](#))

In the end, I decided that degrowth appealed to me because I was like a junkie and degrowth was the dealer providing me with the guilt trip I felt I needed to feel. It was an unhealthy relationship. It didn't make me happy. It didn't offer workable solutions. And most importantly for the debunker I am at heart: it got its facts wrong.

So I put my feelings aside and started calculating and gathering facts. First I did a deep dive into planetary boundaries. I found the problem isn't wealth. The problem isn't energy. These abstractions don't transgress planetary boundaries or cause poverty. The real problems are far more specific.

Raw materials, geopolitical tensions, land use, and pollution. Those are the things we need to degrow. These are the real concerns. They deserve serious examination. They are what the coming chapters of the book are about. And when you examine them, they dissolve. Not because they're unimportant, but because the solutions are better than we thought.

Let's start with raw materials. What if I told you we could use give every human on earth abundant energy and *still* use a *thousand* times less raw material?

8.2 Coal Is Consumed, Sunlight Is Forever

Fossil fuels are consumed. Every kilowatt-hour requires new fuel to be mined, transported, and burned. The material passes through once and is gone forever, transformed into carbon dioxide and ash. You cannot recycle exhaust. The coal that powered last year's electricity is now dispersed in the atmosphere.

Solar panels work on sunlight that's produced fresh daily. Solar panels themselves aren't used up. They sit on your roof for thirty years, converting photons to electrons. When the panel eventually gets a lower efficiency, you can take it apart and recover the components.

That's the ideal, but we live in a less than ideal world. So here are the realistic numbers.

To produce one kilowatt-hour of electricity with coal, you have to burn approximately 500 grams of coal. ([Global Energy Monitor, 2025](#)) That coal is gone. It also produces carbon dioxide as it burns by combining the carbon atoms with oxygen atoms. Every 500 grams of coal produces roughly one and a half kilos of carbon dioxide that is dispersed through the atmosphere. ([Whitaker, 2012](#))

To produce one kilowatt-hour with a complete solar system, panels plus batteries plus power converter, you need approximately 2 grams of material spread across the lifetime of

the equipment. That's already 250 times less. Before any recycling. Before any future improvements. Right now. That's not a policy standpoint. It's a physical fact.

But we can do much better. Those 2 grams of material don't vanish when the panel reaches end of life. The emissions are also not as dispersed as tailpipes and chimneys. We could bring the solar panels to recycling addresses when they reach end of life after 35 years of so. With 95% recycling, already achievable for most components, you need only 0.1 grams of new material from mines per kilowatt-hour. 500 grams versus 0.1 grams. That's 5,000 times less raw material. Pause on that. Five thousand times. That's not an incremental improvement. That's not something you achieve by preaching and talking people into feeling guilty. Five thousand times less is an entirely different category of thing.

The bathtub metaphor helps me to understand this. From the perspective of consumers, coal isn't just a stock we're draining: it's flow we keep consuming continuously. Every kilowatt-hour requires new material through the system that is burned and gone at the end. But silicon, copper, aluminum? These are stocks we're building. The atoms don't vanish when used. They stay put, available for the next cycle. Coal flows through and disappears. Metals accumulate and persist.

With solar and proper recycling, we could produce three and a half times more energy than we use today while extracting a thousand times less material from the Earth. This is effective degrowth. Degrowth of something that matters. Not asking billions to sacrifice. Building systems so efficient that the question of sacrifice doesn't arise. It still *feels* too good to be true for me, but here's where the facts lead me.

Sidebar: The 500 Grams vs 0.1 Grams

How much raw material does it take to produce one kilowatt-hour of electricity?

Electricity from	Input (g/kWh)	Raw material cost (cents/kWh)	CO ₂ output (g/kWh)
Coal	500	5	1,500
Natural gas	200	10	500
Diesel generator	220	17	1,270
Solar system	2	0.4	0
...with 95% recycling	0.1	0.2	0

Sources: Coal consumption and efficiency from ([Global Energy Monitor, 2025](#)). Coal CO₂ lifecycle from ([Whitaker, 2012](#)). Natural gas from ([Quaschnig, 2022](#)) and ([CATF, 2024](#)). Diesel from ([Gordon, 2015](#)). Coal price from ([Energy Institute, 2025](#)).

For fossil fuels, I've counted only the fuel itself. Note the CO₂ column: burning one kilogram of coal produces roughly three kilograms of CO₂ because the carbon combines with oxygen from the air.

For solar, I've included everything needed for a standalone system: the panels themselves, the batteries for storage, and the power converter.

Without recycling, solar uses 250 times less material than coal.

With 95% recycling, solar uses 5,000 times less new material from mines.

This dematerialization explains why the learning curve that we already talked about can be so steep for solar. When raw materials dominate your costs, improvement is constrained. You can optimize mining, but digging is a mature technology. With coal, raw material costs are about 5 cents per kilowatt-hour. That's a floor you can't get under.

When materials are a tiny fraction of lifetime energy output, learning can compound indefinitely. Solar's raw material cost is heading toward 0.2 cents per kilowatt-hour, and it can go lower still. Dematerialization is the engine of the learning curve.

8.3 What's Actually Inside

Critics talk about "critical minerals" as if solar panels were made of rare and exotic substances. The reality is more mundane.

A solar panel is mostly glass. About two-thirds by weight. That's sand, essentially, processed into sheets. ([Fraunhofer ISE, 2025](#))

The frame is aluminum, about 13%. ([Fraunhofer ISE, 2025](#)) Aluminum is the third most abundant element in Earth's crust.

Then comes the plastic encapsulation of the cells and the backsheet, about 10%. ([Fraunhofer ISE, 2025](#)) The silicon that actually converts light to electricity is only about 2.5% of the panel's weight. Glass and silicon are made from SiO_2 , more commonly known as "sand."

Copper for wiring makes up about 1%. Silver appears in trace amounts that are being replaced by copper.

More than 90% of a solar panel, by weight, is made from sand, aluminum and plastic. Not exactly scarce resources. ([Fraunhofer ISE, 2025](#))

Sidebar: Materials in a 10 kWp Solar System

A 10 kWp (kilowatt-peak) solar system can power a typical European home, including heating and an electric vehicle.

Solar Panels (10 kWp)

Material	Weight (kg)	Cost (\$)
Glass	280	85
Aluminum (frame)	53	139
Polymers (EVA + backsheet)	43	50

Material	Weight (kg)	Cost (\$)
Silicon	11	28
Copper	5	52
Other	23	42
Total	~415	~396

Material weights based on ([Fraunhofer ISE, 2025](#)) composition data for standard crystalline silicon modules.

LFP Battery (10 kWh)

Material	Weight (kg)	Cost (\$)
Steel casing	15.0	9
Graphite anode	11.1	9
Iron	6.0	5
Phosphorus	4.0	4
Copper	2.0	32
Lithium	0.8	40
Other	26	15
Total	~63	~114

Battery composition from ([IRENA, 2024](#)).

Glass and aluminum dominate by weight. The "critical minerals" everyone worries about are small fractions. The total raw material cost is around \$765 for a system that will generate electricity for 30 years.

Batteries tell a similar story. The chemistry that's winning, LFP or lithium iron phosphate, uses iron and phosphorus as its main active materials. Not cobalt. Not nickel. Iron and phosphorus.

A 10 kilowatt-hour LFP battery contains about 6 kilograms of iron, 4 kilograms of phosphorus, 11 kilograms of graphite (carbon), and less than 1 kilogram of lithium. ([IRENA, 2024](#)) The lithium that makes headlines is a little over 1% of the battery pack weight. ([IRENA, 2024](#))

Glass, silicon, aluminum, iron, phosphorus, carbon. These are not exotic substances. When people talk about resource scarcity in renewables, the most discussed materials are lithium, cobalt, rare earth metals, and copper.

8.4 Lithium: The "New Oil" That Isn't

Of all the materials critics worry about, lithium gets the most attention. "White gold," they call it. The new oil. The resource that will determine who wins the twenty-first century.

I understand why. Electric vehicles run on lithium batteries. So do grid storage systems. Demand is growing at about 30 percent per year. ([Benchmark, 2024](#)) If lithium was truly scarce, we might have a problem.

But lithium is not scarce. Not even close.

As of 2025, the world has identified about 115 million tonnes of lithium resources. ([USGS, 2025](#)) Current annual production is about 240,000 tonnes. ([USGS, 2025](#)) At this rate, identified resources would last almost 500 years. And "identified" is the key word. Every time production increases, exploration increases. New deposits are found in new places: Germany, Canada, Mexico, the United States, across Africa. The reserves grow alongside the demand. I did a deep dive recently and found a whopping 95% correlation between production and reserves. And this is before we start talking about getting it from sea water where it is truly abundant. So the idea that it is scarce is ludicrous.

On top of this there are alternatives. In April 2025, CATL, the world's largest battery maker, launched the first mass-produced sodium-ion battery for electric vehicles. ([CATL, 2025](#)) Sodium sits right below lithium on the periodic table, behaves similarly in batteries, and is over a thousand times more abundant in Earth's crust. ([Adelhelm, 2015](#)) It is literally the main ingredient in table salt. Sodium-ion batteries store a bit less energy per kilogram but CATL's newest sodium batteries already achieve 175 watt-hours per kilogram, approaching lithium iron phosphate territory. ([CATL, 2025](#)) For city cars, delivery vans, and stationary storage, this is more than enough.

None of this means lithium mining is unproblematic. In Chile's Atacama Desert, the world's second-largest lithium producer, ([USGS, 2025](#)) mining companies pump brine from beneath ancient salt flats and let it evaporate in enormous ponds. Now it's not that anything lives in the salt flats where this happens. But the dropping water tables lead to land subsiding around extraction wells. Indigenous communities have filed complaints and organized protests. The Chilean government has sued mining companies for exceeding water extraction limits. ([Meckievi, 2025](#)) For that reason, I'm glad that production has shifted to countries like Australia that now produces more lithium than any other country. ([USGS, 2025](#)) Hard rock mining generates tailings and requires significant energy, but this energy increasingly comes from solar panels, there is no water scarcity, and Australia has strong environmental regulations. Personally, I think the future is filtering brine from deep wells and sea water that I expect will become both easiest and most sustainable.

But the question is not whether lithium mining is perfect. Mining never is. The question is whether lithium scarcity will block the energy transition. And the answer is no. We already know where to find enough for centuries, there are large deposits on every continent, and sodium batteries could take over if lithium ever became a problem.

Lithium is not the new oil. Oil is geologically concentrated in a few places and must be burned continuously. Lithium is found nearly everywhere, is used in durable goods that last for decades, and can be recycled indefinitely. There will be no lithium-OPEC. There cannot be. The physics and geology are fundamentally different.

8.5 Cobalt: The Problem We Routed Around

Cobalt presents a different kind of challenge. It is not about scarcity. It is about where cobalt comes from and how it is mined.

The Democratic Republic of the Congo produces about three-quarters of the world's cobalt. (USGS, 2024) Much of that comes from large industrial mines. But perhaps 15 to 30 percent comes from what are called artisanal mines: small operations where families dig by hand, sometimes in tunnels that collapse, sometimes with children doing the digging. (Sovacool, 2019) UNICEF estimates that 40,000 children work in Congolese cobalt mines, some as young as six. (UNICEF, 2014) The conditions are brutal: respiratory diseases from dust, spinal deformities from carrying heavy loads, the constant risk of being buried alive.

This is a genuine horror. Anyone who cares about human rights should care about what happens in these mines. Anyone buying an electric vehicle or a smartphone should at least be aware of where the materials might have come from.

But there is something worth noting. Cobalt has been used in petroleum refining for decades. Every oil refinery in the world uses cobalt-molybdenum catalysts to remove sulfur from diesel and gasoline. This has been standard practice since the 1950s. (Topsøe, 1996) Nobody objected. Nobody made documentaries. Nobody called for boycotts. Cobalt was invisible because it was embedded in the fossil fuel system that everyone took for granted.

Then cobalt appeared in electric vehicle batteries, and suddenly it became a reason to reject the energy transition altogether.

Here is what actually happened while critics were pointing fingers. The battery industry routed around the problem. Not through ethics alone, but through chemistry.

Lithium iron phosphate batteries, LFP for short, contain zero cobalt. They use iron and phosphorus instead: two of the most abundant and inexpensive elements on Earth. In 2020, LFP batteries held about 10 percent of the global electric vehicle market. (IEA, 2024); (McKinsey, 2024) By 2024, they held nearly 40 percent. (IEA, 2025) In China, where the transition is furthest along, LFP dominates with over three quarters of market share. (IEA, 2025) Tesla uses LFP in its standard-range vehicles. (Lambert, 2021) BYD, the world's largest electric vehicle maker, (SCMP, 2025) has built its empire on cobalt-free batteries.

The trend is accelerating. Industry analysts expect LFP to become the dominant battery chemistry by 2028, (Wood Mackenzie, 2022) not because companies have suddenly become ethical, but because iron and phosphorus are cheaper than cobalt and nickel, and because LFP batteries last longer and are safer. The market is solving the problem that critics said was unsolvable.

Even the lightweight lithium batteries that still use cobalt, so called NCM batteries for their use of Nickel, Cobalt and Manganese, are using less of it. A decade ago, cathodes were roughly one-third cobalt. (Buchmann, 2025) Today's high-nickel formulations have reduced

that to under 10 percent, and new chemistries are pushing toward 5 percent. (Conrad, 2024)

Does this mean we can ignore the conditions in Congolese mines? Of course not. The International Labour Organization, UNICEF, and various industry initiatives are working to formalize artisanal mining, get children into schools, and improve safety conditions. These efforts matter. But they exist alongside a technological shift that is making the problem smaller.

In absolute terms, cobalt is not scarce either. Global resources are estimated at 25 million tonnes, (USGS, 2024) enough for centuries of production at current rates. Indonesia has emerged as a major new producer, going from almost nothing in 2021 to 10 percent of global output in 2024. (Mining Technology, 2025) Australia, Canada, Cuba, and the Philippines all have significant deposits. The concentration in the Congo is a result of history and economics, not geology.

8.6 Rare Earth Metals: Neither Rare Nor Earthy

Whoever named them should be fired from the naming committee. "Rare earth metals" sounds like something precious and scarce, hoarded in secret vaults. In reality, these elements are neither rare nor earths. They were called "rare" because they were first isolated from uncommon minerals, not because they are geologically scarce. Cerium, the most common rare earth element, is more abundant in Earth's crust than copper. (USGS, 2024) The "earth" part is an archaic term for metal oxides. The name stuck, and it has been causing confusion ever since.

Rare earth elements are used primarily in permanent magnets. These magnets appear in wind turbines, electric vehicle motors, and various electronic devices. And although they are not particularly scarce, they are often the most expensive raw material in an electric motor. For that reason, engineers are increasingly making them optional.

For example by using synchronous reluctance motors that use no permanent magnets at all and are becoming increasingly popular in industrial applications. These motors are a bit harder to control and design, but as you will see in the second part of this book our ability to control electricity is increasing at an impressive pace. Others choose to use permanent magnets that do away with the rare earths. Tesla announced in 2023 that its next-generation motors would eliminate rare earth elements entirely, using alternative magnet designs. (Tesla Investor Day, 2023) Other manufacturers are following. The transition from rare-earth-dependent to rare-earth-optional is already underway.

The pattern should be familiar by now. Critics identify a potential bottleneck. Engineers route around it. The bottleneck never materializes, but it stays a talking point for many years.

8.7 Copper: The Tightest Constraint

When I researched this with master students, copper was the one that gave us pause. This is a concern for many other experts too and that's easy to understand. Your house contains about 90 kilograms of copper wiring. (CDA, n.d.) An electric vehicle needs 80 to 90 kilograms, about three and a half times what a combustion car uses. (Wood Mackenzie, 2019) A 3-megawatt wind turbine contains up to 4.7 tonnes. (CDA, 2018) The grid that connects everything runs on copper. As we electrify transport, heating, and industry, demand is surging.

The numbers look daunting. Global demand reached 27 million tonnes in 2024 and is projected to hit 37 million tonnes by 2050. (IEA, 2025) The IEA forecasts a 30 percent supply shortfall by 2035. (IEA, 2025) Meanwhile, ore grades have declined roughly 40 percent since 1991. (IEA, 2025; Calvo et al., 2016) The world's largest copper mine, Escondida in Chile, started in 1990 processing ore at 2.5 percent copper. Today it's at 0.5 percent, digging five times as much earth for the same output. (Earth Resource Investments, 2025; BHP, 2024) Experts and headlines scream shortage. What I found was different.

The U.S. Geological Survey reports current reserves at about 1,000 million tonnes. (USGS, 2025) At 2024 production rates, that's roughly 40 years of supply. Sounds tight. But since 1960, there have always been about 40 years of reserves. (ICSG, 2023) Every decade, people predict we'll run out in 40 years. Every decade, reserves grow alongside production. Between 2000 and 2022, the world mined 396 million tonnes of copper. Over the same period, reserves *increased* by 457 million tonnes. (ICSG, 2023) We ended up with more than we started with.

Beyond reserves lie identified resources totaling 2.1 billion tonnes, plus another 3.5 billion tonnes that geological surveys predict exist based on where copper forms. (USGS, 2025; Johnson et al., 2014) At current consumption, identified resources alone would last over 200 years.

The real question is whether we can extract it fast enough. Mining projects take 15 to 20 years from discovery to production. (IEA, 2021) Investment has lagged. This is a legitimate concern for the 2030s.

But two forces are solving it: substitution and recycling.

Aluminum has already replaced copper in most power transmission lines worldwide. Those high-voltage cables strung between pylons? Almost entirely aluminum. It's only 60 percent as conductive as copper, but one-third the weight and one-third the price. (USGS, 2025) Engineers are pushing aluminum further: underground cables, wind turbine generators, vehicle wiring. Tesla's next-generation vehicles will use aluminum where possible and run at 48 volts instead of 12, cutting conductor needs by three-quarters. (Musk, 2023)

The pattern should be familiar. When a material gets tight, engineers route around it. Sodium instead of lithium. LFP instead of cobalt. Reluctance motors instead of rare earths. Aluminum instead of copper.

Then there's recycling. Copper may be the oldest recycled material in civilization. We've been melting it down for 10,000 years. The atoms in your wiring might once have been Bronze Age tools or Roman plumbing. An estimated 80 percent of all copper ever mined is still in use, locked in buildings and infrastructure. (ICA, 2024) It hasn't been consumed; it's been accumulated. As today's infrastructure ages out, it becomes a copper deposit for future generations.

The 2030s may see genuine supply crunches and price spikes. This is not a problem that solves itself without effort. But it solves itself *with* effort. Aluminum substitution is proven. Recycling infrastructure is growing. And the fundamental resource base is not scarce: it's a matter of extraction rates, not geology.

Most importantly, copper shares the characteristic that distinguishes all renewable materials from fossil fuels. It's a stock, not a flow. Once we've built the renewable energy system, the copper stays in place, generating electricity for decades, available for recycling when eventually replaced. We're not burning through it; we're building with it. Which brings us to recycling. Because isn't that a scam?

8.8 Recycling: This Time Is Different

You might be skeptical about recycling promises. You should be. We've heard them before. Plastic recycling turns out to be largely a fraud: less than 10% of plastic ever produced has actually been recycled. (Geyer, 2017) The little triangle on your yogurt container is closer to greenwashing than reality. So when I tell you that solar panels and batteries can achieve 95% recycling rates, you're right to ask: why should I believe you?

I have an answer, and it's sitting under the hood of your car. Lead-acid batteries have achieved 99% recycling rates for decades. (EPA, 2020) Not a target. Not a promise. The measured reality, confirmed every year. We already know how to do this. We've been doing it for a century. Lead-acid batteries have higher recycling rates than aluminum cans, glass bottles, or newspapers. (CSS, 2025) A typical new lead-acid battery contains 80% recycled material. (EPA, 2020) The lead has been cycling through batteries for decades, perhaps longer.

This matters because the objection I hear most often about electric vehicles and grid storage is: "What happens to all those batteries?" The worry is lithium batteries piling up in landfills, toxic and wasted. But we already recycle batteries at industrial scale. We've been doing it for a century. The question isn't whether battery recycling is possible. The question is whether the conditions that made lead-acid recycling work also apply to lithium batteries and solar panels.

8.9 What Makes 99% Recycling Possible

Lead-acid batteries didn't reach their 99% recycling rate by accident. Four conditions had to align, and understanding them tells us what to expect for solar and lithium.

First, the materials must be valuable. Lead commands a decent price. When a battery reaches end of life, someone will pay to collect it because there's money in the lead. Collection happens because it's profitable. This is very different from plastic where new plastic is actually cheaper than recycled plastic and both cost very little. With plastic the business simply isn't there.

Second, the products are consistent. Every lead-acid car battery has roughly the same design: lead plates, sulfuric acid electrolyte, plastic casing. Recyclers know exactly what's coming in. They can build efficient processes optimized for materials they'll encounter every time. Compare this to plastic recycling, where a single collection bin might contain dozens of different polymers, often contaminated with food and other non-plastics, mixed in ways that make separation expensive and imperfect.

Third, the physics allows it. Lead is an element. You can melt it down and recast it indefinitely without degradation. The lead in your car battery might once have been in a Roman pipe. It might become part of a battery a century from now. Metals don't wear out the way polymers do. Plastic chains break down with each recycling cycle; after a few rounds, the material is only good for park benches. There are ways to get around this and upcycle plastic but they are uneconomical. Metals like lead on the other hand, never run out of cycles.

Fourth, the infrastructure exists. This took decades to build. A network of collection points, processing facilities, regulations requiring retailers to accept old batteries. The system works because someone designed it to work. And to be honest, this is why we shouldn't expect 99% recycling for solar panels and lithium batteries tomorrow. The collection networks don't exist yet. Not just because everything is so new, but also because there is little to recycle yet. Most of the solar panels and car batteries that were made will take over a decade before they are ready for recycling. Building the infrastructure takes time and volume.

But the first three conditions, value, consistency, and physics, are already in place, so we can be confident the infrastructure will be built.

8.10 Renewables are Great for Recycling

Solar panels contain roughly \$10 to \$15 worth of recoverable materials: silver, copper, high-purity silicon, aluminum, glass. (Tao, 2020) That sounds modest until you realize there will be hundreds of millions of panels reaching end of life every year by 2050. Rystad Energy estimates the annual value of recyclable materials from solar panels could approach \$80 billion by 2050. (Rystad, 2022) Companies are racing to build recycling capacity because there's real money in it.

Lithium batteries offer even better economics. A single electric vehicle battery pack contains hundreds (with lithium iron phosphate batteries) or even thousands of dollars (with nickel cobalt manganese batteries) worth of recoverable metals. (Ma, 2024) Unlike

plastic, which costs more to recycle than to replace, these materials command prices that make collection profitable.

The consistency is there too. Solar panels are manufactured products with standard formats. Every crystalline silicon panel has the same basic structure: glass on top, encapsulant layer, silicon cells, backsheet, aluminum frame. The composition is known and predictable. Recyclers can optimize each step because they know what's coming in. Lithium batteries are similarly consistent and especially the larger cells now becoming standards give recyclers nice hefty chunks of battery to work with.

And the physics is on our side. Aluminum is aluminum. Copper is copper. Silicon can be purified back to solar grade. You can melt these materials and reuse them indefinitely. We've been recycling copper for 5,000 years. (CDA, 2024) An estimated 80% of all copper ever mined in human history is still in use somewhere today, locked up in buildings, wires, and equipment. (ICA, 2024) It hasn't been consumed. It's been accumulated. You cannot destroy the atoms in metals. Recycling just means that you relocating atoms into new useful configurations. This is why the comparison between fossil fuel extraction and renewable mining is misleading. Fossil fuels are consumed. Renewables are a stock that we are building.

8.11 What's Already Happening

The infrastructure is being built faster than most people realize.

First Solar has been recycling its thin-film panels since 2005. (Semiconductor Today, 2024) Twenty years of industrial-scale recycling, not a pilot project. They recover over 90% of semiconductor material and 90% of glass. Their facilities in the U.S., Germany, India, Malaysia, and Vietnam process 2.6 million modules per year. In 2023, their global material recovery rate averaged 95%. They've recycled nearly 400,000 metric tonnes of panels to date. (Semiconductor Today, 2024)

For lithium batteries, Redwood Materials has become the largest recycler in North America. (Bloomberg, 2024) Founded by Tesla's former chief technology officer, (Wikipedia, 2025) they now receive over 20 gigawatt-hours of batteries annually. (Redwood Materials, 2025) They claim 95% recovery of lithium, cobalt, nickel, and copper. (Redwood Materials, 2025) In 2024 alone, they recycled enough material to supply batteries for 1.5 billion phones. (Redwood Materials, 2025)

The recovery rates for individual materials have improved dramatically in the past five years. Silicon extraction has improved dramatically, with recent processes achieving over 90% recovery rates. (Wang, 2024); (Sharma, 2022) Silver recovery, once considered too difficult, now reaches over 95% with the resulting silver being over 98% pure, with some methods completing extraction in under ten minutes. (Zante, 2022); (Shen, 2023) Glass and aluminum that make up most of the material in a solar panel routinely hit 90 to 95%. (EPA, 2025)

These aren't laboratory demonstrations. They're operating facilities processing real waste streams at industrial scale.

8.12 What We Can Honestly Expect

I don't expect solar panels and lithium batteries to reach 99% recycling rates any time soon. Lead-acid had a century head start and the advantage of being concentrated in auto shops rather than scattered across millions of rooftops. Collecting distributed equipment is harder. But 90% is achievable, even in the short term, because many facilities are already there.

With 90% recycling, the material extracted from mines drops by a factor of ten compared to no recycling. With 95% recycling, already achieved for some components, it drops by a factor of twenty. The 500 grams of coal burned per kilowatt-hour versus 0.1 grams of new material extracted with solar recycling: that comparison holds even if we never reach lead-acid's 99%.

The trajectory matters too. Recovery rates that seemed impossible five years ago are now routine. Processes that took hours now take minutes. The learning curve applies to recycling just as it applies to manufacturing. Every year, the processes get better and the economics improve.

Europe already requires solar panel manufacturers to finance collection and recycling under the WEEE directive. (EU, 2012) California has solar-specific recycling regulations. (CA DTSC, 2021) The pattern is spreading. These are the same policy tools that built the lead-acid recycling infrastructure over the past century, now being applied to renewable equipment.

By 2050, an estimated 78 million tonnes of solar panels will reach end of life globally. (IRENA, 2016) That's an enormous waste stream if we get this wrong. It's also an enormous resource stream if we get it right: enough recovered material to manufacture tens of millions of new panels every year without new mining.

The lithium battery recycling market is expected to grow from approximately \$5 billion in 2024 to over \$24 billion by 2032. (Fortune, 2024) Redwood Materials is building toward 500 gigawatt-hours of annual recycling capacity by 2030, enough to supply materials for 5 million electric vehicles. (Straubel, 2022) These are investments being made now, facilities being built now.

We know how to do this. We've proven it with lead-acid for a century. The conditions that made lead-acid recycling work, value, consistency, physics, also apply to solar panels and lithium batteries. The infrastructure is being built. The question isn't whether recycling will happen. It's how fast we build the systems to make it happen at scale.

8.13 Spending the Inheritance vs Guaranteed Income

From fossil fuels to renewables is a fundamental shift in the mental model. They aren't called renewables for nothing. Fossil fuels is like spending an inheritance. We discovered

millions of years of stored sunlight, compressed into coal and oil and gas. We've been withdrawing from that bank account for three centuries. We draw it down a million times faster than it was created. The spending has been glorious, lifting billions from poverty, building modern civilization. But the account is finite. The balance is dropping. The end is coming into view quickly.

Renewables is like getting a regular income. The sun deposits energy into your account every day. You live on what arrives. And not just while you're employed, or until your pension runs out. For billions of years. The materials you use to capture that deposit last for decades and can be reused again and again.

Solar and wind are not something strange and new. In a way, it's going back to a past in which we lived on what the earth gave us. The sun showed up daily, growing crops, filling rivers, pushing sails. We spent what arrived. Burning wood was just a very inefficient way to collect sunlight, and now we're replacing it with something more efficient. Same source. Same principle. We're just finally getting good at it.

8.14 The Big Picture

Sometimes it helps to step back and see the big picture in all its glory.

Global fossil fuel extraction runs about 15 billion tonnes per year. (UNEP, 2019) That's coal plus oil plus gas. This isn't a one-time cost during some transition period. This is every year for as long as fossil fuels last.

The total material needed to build a fully renewable global energy system, all the solar panels, all the wind turbines, all the batteries, all the power converters, comes to roughly 6 to 7 billion tonnes cumulatively through 2050. And 90% of that is steel and aluminum. (ETC, 2023)

After that, recycling dominates. The system maintains itself with minimal new extraction.

So: fossil fuels require 15 billion tonnes extracted every year, forever. Renewables require 6 to 7 billion tonnes total over 25 years. The renewable system requires less material to build than the fossil system extracts in a single year. And then we start to recycle and no physical law says that we can't recycle everything in the distant future.

People miss this because we've normalized fossil extraction. Coal trains and oil tankers are invisible infrastructure. They've been running for so long that we don't see them anymore. Renewable construction is new and visible. A wind farm going up catches your eye. A coal mine operating for the hundredth year doesn't. The fish doesn't notice the water. Even when it's running out.

8.15 Dematerialization Continues

Everything I've described is based on today's technology. The trajectory points toward even less material per kilowatt-hour.

Silicon wafers in solar panels keep getting thinner. Each generation uses less silicon than the last. Newer panel designs are eliminating silver entirely, replacing it with copper or aluminum contacts.

Battery energy density keeps improving. The same kilowatt-hour of storage requires less material every year. Sodium-ion batteries are entering production, eliminating lithium from the equation entirely for stationary storage.

Power converters are approaching efficiencies of 100%. This means almost no waste heat, which in turn means incredibly compact and lightweight devices that uses very little energy.

Panel lifetimes keep extending. Some panels installed in the 1970s still generate electricity today, albeit with some degradation. ([Bellini, 2021](#)) Forty-year lifetimes are becoming plausible. Fifty years may follow.

On top of this recycling is developing fast. Recovery rates that seemed impossible five years ago are now routine. Processes that took hours now take minutes. Technologies that required toxic chemicals now use environmentally friendly alternatives. So if today we are at 0.1 grams per kilowatt-hour with recycling, this could be 0.05 grams or less by 2050. Dematerialization is not a one-time achievement. It's an ongoing process, driven by the same learning curves we discussed earlier.

Fossil fuel is the only truly scarce material in the energy transition. "Critical minerals" aren't scarce and can be recycled endlessly. Dematerialization is the most effective form of degrowth. I now feel less guilty about the luxury I live in because I know my flourishing doesn't require anyone's deprivation. But could minerals make us dependent on new masters? That's the next question.